

Неделя 11 Канонические преобразования

задача 6

Дано: $\tilde{q} = p \exp q$, $\tilde{p} = q + \exp(-q) + \ln p$

Найти: γ, F

Решение:

Критерий каноничности: $\tilde{p} \delta \tilde{q} - \gamma p \delta q = -\delta F$

$$\tilde{p} \delta \tilde{q} - \gamma p \delta q = (q + e^{-q} + \ln p) \delta(p e^q) - \gamma p \delta q = (q + e^{-q} + \ln p)(p e^q \delta q + e^q \delta p) - \gamma p \delta q =$$
$$= (q p e^q + p + p e^q \ln p - \gamma p) \delta q + (q e^q + 1 + e^q \ln p) \delta p = -\delta F(q, p)$$

$$\left. \begin{aligned} -\frac{\partial F}{\partial p \partial q} &= q e^q + 1 - \gamma + e^q (\ln p + 1) \\ -\frac{\partial F}{\partial q \partial p} &= e^q + q e^q + e^q \ln p \end{aligned} \right\} \text{ - равны при } \gamma = 1.$$

$$-\frac{\partial F}{\partial p} = q e^q + 1 + e^q \ln p \Rightarrow -F = \int (q e^q + 1 + e^q \ln p) \delta p = q p e^q + p + e^q \int \ln p \delta p + c(q) =$$
$$= q p e^q + p + e^q p (\ln p - 1) + c(q) =$$

$$-\frac{\partial F}{\partial q} = p e^q (q + 1) + e^q p (\ln p - 1) + \frac{dc}{dq} = q p e^q + p e^q \ln p + \frac{dc}{dq} = q p e^q + p e^q \ln p \Rightarrow$$
$$\Rightarrow \frac{dc}{dq} = 0 \text{ Возьмем } c = 0$$

$$\text{Тогда } F = - (p e^q (q + \ln p - 1) + p) + \varphi(t)$$

Ответ: $\gamma = 1$, $F(q, p) = - (p e^q (q + \ln p - 1) + p) + \varphi(t)$, φ - любая

задача 30

Дано: $\tilde{q}_i = q_i$, $\tilde{p}_i = \gamma p_i - \frac{\partial \varphi(\vec{q}, t)}{\partial q_i}$, φ - непрерывно дифференцируема, $\gamma = \text{const}$

Показать, что преобразование каноническое, найти c и F , $\tilde{H}$

Решение:

Критерий каноничности: $\tilde{p}^T \delta \tilde{q} - c \tilde{p}^T \delta q = -\delta F(\vec{q}, \vec{p}, t)$

$$\tilde{p}^T \delta \tilde{q} - c \tilde{p}^T \delta q = \sum_{i=1}^n \left(\left(\gamma p_i - \frac{\partial \varphi}{\partial q_i} \right) \delta q_i - c p_i \delta q_i \right) = \sum_{i=1}^n \left[(\gamma - c) p_i - \frac{\partial \varphi}{\partial q_i} \right] \delta q_i = -\delta F$$

$$-\frac{\partial F}{\partial p_i} = 0 \Rightarrow F = F(\vec{q}, t)$$

$$-\frac{\partial F}{\partial q_i} = (\gamma - c) p_i - \frac{\partial \varphi(\vec{q}, t)}{\partial q_i} \text{ - не зависит от } p_i \text{ при } c = \gamma.$$

$$\text{Тогда } \frac{\partial F}{\partial q_i} = \frac{\partial \varphi}{\partial q_i} \Rightarrow F(\vec{q}, t) = \varphi(\vec{q}, t)$$

$$\tilde{H}(\vec{q}, \vec{p}, t) = c H(\vec{q}, \vec{p}, t) + \frac{\partial F(\vec{q}, \vec{p}, t)}{\partial t} + \tilde{p}^T \frac{\partial \tilde{q}}{\partial t} = \gamma H\left(\vec{q}, \frac{1}{\gamma} \left(\vec{p} + \frac{\partial \varphi(\vec{q}, t)}{\partial \vec{q}} \right), t\right) + \frac{\partial \varphi(\vec{q}, t)}{\partial t}$$

Ответ: $c = \sigma$, $F(\vec{q}, t) = \psi(\vec{q}, t)$, $\tilde{H}(\vec{q}, \vec{p}, t) = \sigma H(\vec{q}, \frac{1}{\sigma}(\vec{p} + \frac{\partial \psi}{\partial \vec{q}}), t) + \frac{\partial \psi}{\partial t}$

223.47

Дано: $H(q, p) = \frac{1}{2}(p^2 + \omega^2 q^2)$, $H_1(\tilde{q}, \tilde{p}) = \frac{1}{2}\omega(\tilde{p}^2 + \tilde{q}^2)$

Построить каноническое преобразование, переводящее H в H_1 , найти γ и F .

Решение:

$$\tilde{H}(\tilde{q}, \tilde{p}, t) = \sigma H(q, p, t) + \frac{\partial F}{\partial t} + \tilde{p} \cdot \frac{\partial \tilde{q}}{\partial t}$$

$$\frac{1}{2}\omega(\tilde{p}^2 + \tilde{q}^2) = \sigma \cdot \frac{1}{2}(p^2 + \omega^2 q^2) + \frac{\partial F}{\partial t} + \tilde{p} \frac{\partial \tilde{q}}{\partial t}$$

Пусть $\tilde{q} = \tilde{q}(q, p)$, $F = F(q, p)$, тогда

$$\omega \tilde{p}^2 + \omega \tilde{q}^2 = \sigma p^2 + \sigma \omega^2 q^2$$

Выберем $\tilde{p} = \alpha p$, $\tilde{q} = \alpha \omega q$ ($\sigma = 4\omega$, выбираем любое)

Критерий каноничности: $\tilde{p} \delta \tilde{q} - \sigma p \delta q = -\delta F$

$$\tilde{p} \delta \tilde{q} - \sigma p \delta q = 4\omega p \delta q - 4\omega p \delta q = 0 = -\delta F \Rightarrow F = \text{const}$$

Ответ: $\tilde{q} = 2\omega q$, $\tilde{p} = 2p$, $\gamma = 4\omega$, $F = \text{const}$

223.102

Выяснить, является ли каноническим с помощью критерия

в $(q, \tilde{q})$ описании, найти c и S :

$$\tilde{q}_1 = \frac{1}{\sigma p_1 + q_2} - q_1, \quad \tilde{p}_1 = -(\sigma p_1 + q_2)$$

$$\tilde{q}_2 = \frac{2}{\sigma p_2 + q_1} - 2q_2, \quad \tilde{p}_2 = -\frac{1}{2}(\sigma p_2 + q_1)$$

Решение:

Критерий каноничности $(q, \tilde{q})$ описании: преобразование канонично,

если $\exists c, S(\vec{q}, \vec{\tilde{q}}, t)$:

$$\vec{\tilde{p}} = -\frac{\partial S}{\partial \vec{\tilde{q}}}, \quad c\vec{p} = \frac{\partial S}{\partial \vec{q}}$$

Выразим $\vec{p}$ и $\vec{\tilde{p}}$ через $\vec{q}$ и $\vec{\tilde{q}}$:

$$p_1 = \frac{1}{\sigma} \left(\frac{1}{q_1 + \tilde{q}_1} - q_2 \right), \quad p_2 = \frac{1}{\sigma} \left(\frac{2}{2q_2 + \tilde{q}_2} - q_1 \right)$$

$$\tilde{p}_1 = - \left(\frac{1}{q_1 + \tilde{q}_1} - q_2 + q_2 \right) = -\frac{1}{q_1 + \tilde{q}_1}, \quad \tilde{p}_2 = -\frac{1}{2} \left(\frac{2}{2q_2 + \tilde{q}_2} - q_1 + q_1 \right) = -\frac{1}{2q_2 + \tilde{q}_2}$$

$$\frac{1}{q_1 + \tilde{q}_1} = \frac{\partial S}{\partial \tilde{q}_1} \quad (1)$$

$$\frac{1}{2q_2 + \tilde{q}_2} = \frac{\partial S}{\partial \tilde{q}_2} \quad (2)$$

$$\frac{1}{q_1 + \tilde{q}_1} - q_2 = \frac{\gamma}{c} \frac{\partial S}{\partial q_1} \quad (3)$$

$$\frac{2}{2q_2 + \tilde{q}_2} - q_1 = \frac{\gamma}{c} \frac{\partial S}{\partial q_2} \quad (4)$$

$$(1) \Rightarrow S = \ln(q_1 + \tilde{q}_1) + f_1(q_1, q_2, \tilde{q}_2) \quad | \Rightarrow S = \ln(q_1 + \tilde{q}_1) + \ln(2q_2 + \tilde{q}_2) + f(q_1, q_2)$$

$$(2) \Rightarrow S = \ln(2q_2 + \tilde{q}_2) + f_2(q_1, q_2, \tilde{q}_1)$$

$$\frac{\partial S}{\partial q_1} = \frac{c}{\gamma} \left(\frac{1}{q_1 + \tilde{q}_1} - q_2 \right) = \frac{1}{q_1 + \tilde{q}_1} + \frac{\partial f(q_1, q_2)}{\partial q_1}$$

$$\text{Пусть } c = \gamma, \text{ тогда } \frac{\partial f}{\partial q_1}(q_1, q_2) = -q_2 \Rightarrow f(q_1, q_2) = -q_1 q_2 + C(q_2)$$

$$\frac{\partial S}{\partial q_2} = \frac{2}{2q_2 + \tilde{q}_2} + \frac{\partial f(q_1, q_2)}{\partial q_2} = \frac{2}{2q_2 + \tilde{q}_2} - q_1 + C(q_2) = \frac{2}{2q_2 + \tilde{q}_2} - q_1 \Rightarrow C(q_2) = 0$$

Таким образом, найдем $S(\vec{q}, \vec{\tilde{q}}, t) = \ln(q_1 + \tilde{q}_1) + \ln(2q_2 + \tilde{q}_2) - q_1 q_2 + \varphi(t)$ и $c = \gamma$ такие что $\vec{p} = -\frac{\partial S}{\partial \vec{q}}, c\vec{p} = \frac{\partial S}{\partial \vec{q}} \Rightarrow$ преобразование каноническое.

Ответ: каноническое, $c = \gamma, S(\vec{q}, \vec{\tilde{q}}, t) = \ln(q_1 + \tilde{q}_1) + \ln(2q_2 + \tilde{q}_2) - q_1 q_2 + \varphi(t)$

223.110

Дано: μ -вапентность, $S = \sum_{i,j=1}^n a_{ij} q_i \tilde{q}_j = \vec{q}^T A \vec{\tilde{q}}$

Найти: формулы преобразования.

Решение:

Критерий каноничности преобразования в $(q, \tilde{q})$ -описании:

$$\vec{p} = -\frac{\partial S}{\partial \vec{q}}, \mu \vec{p} = \frac{\partial S}{\partial \vec{q}}$$

$$\tilde{p}_k = -\frac{\partial S}{\partial \tilde{q}_k} = \frac{\partial}{\partial \tilde{q}_k} \sum_{i,j=1}^n a_{ij} q_i \tilde{q}_j = \sum_{i=1}^n a_{ik} q_i = (\vec{q}^T A)_{k-\text{ая}} = (A \vec{q})_{k-\text{ая}} \Rightarrow \vec{\tilde{p}} = A \vec{q}$$

$$\mu p_k = \frac{\partial S}{\partial q_k} = \frac{\partial}{\partial q_k} \sum_{i,j=1}^n a_{ij} q_i \tilde{q}_j = \sum_{j=1}^n a_{kj} \tilde{q}_j = (A \vec{\tilde{q}})_{k-\text{ая}} \Rightarrow \mu \vec{p} = A \vec{\tilde{q}} \Rightarrow \vec{\tilde{q}} = \mu A^{-1} \vec{p}$$

Ответ: $\vec{\tilde{q}} = \mu A^{-1} \vec{p}, \vec{\tilde{p}} = A \vec{q}$, где $A = \|a_{ij}\|_{1 \leq i,j \leq n}$